HAMZA KHALEM

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience developing, customizing, and maintaining scalable web applications and enterprise software solutions. Strong background in backend development, database management, RESTful APIs, and system integrations using Python, Django, PHP, and PostgreSQL. Experienced in building maintainable software, optimizing application performance, troubleshooting production issues, and developing solutions for large-scale platforms serving over 300,000 users. Holds a Master's degree in Intelligent Computer Systems Engineering.

PROFESSIONAL EXPERIENCE

Software Engineer | Naseej

May 2022 – February 2026

- Developed and maintained enterprise-level e-learning applications based on Moodle, supporting over 300,000 users and up to 25,000 concurrent users.
- Customized and extended Moodle core functionalities and developed plugins, blocks, authentication, enrolment, messaging, reporting, and repository integrations.
- Designed and implemented REST APIs and backend services to support integrations with external systems.
- Developed AI-powered platform features, including AI Tutor, AI Question Creator, and AI-based proctoring capabilities.
- Integrated payment gateways, SMS providers, and third-party authentication services.
- Worked with PostgreSQL, Redis, PHP 8.1, JavaScript, jQuery, and Moodle architecture.
- Improved application performance through database query optimization, caching, and backend enhancements.
- Supported deployment and maintenance workflows using Docker, Rancher, and Azure DevOps pipelines.
- Investigated and resolved application bugs and production issues while ensuring system stability and maintainability.

Software Engineer | Gen42

March 2022 – May 2022

- Developed and maintained web applications for banking, commercial, and business clients.
- Contributed to the development and maintenance of Banque Extérieure d'Algérie's website.
- Worked on the official Samsung Algeria website and product landing pages.
- Developed web interfaces and application features using Django, React, Laravel, and React Native.
- Collaborated on backend and frontend development for internal software projects.

No-Code Engineer | Drakaap

January 2022 – March 2022

- Contributed to the development and maintenance of French government websites.
- Implemented website features and optimized existing application interfaces.

Software Developer | IT Creative

December 2018 – September 2019

- Developed C#/.NET applications for educational institutions.
- Implemented software features for educational management systems.
- Maintained and improved existing application functionality.

TECHNICAL SKILLS

Programming Languages: Python, PHP, JavaScript, C#, SQL.

Backend Development: Django, Django REST Framework, PHP, RESTful APIs.

Frontend Development: React, React Native, jQuery, HTML5, CSS3, Bootstrap.

Databases: PostgreSQL, SQL, Redis.

Software Engineering: Object-Oriented Programming, API Design, Debugging, Performance Optimization, Software Maintenance, Scalable Architecture, Unit Testing.

DevOps & Tools: Docker, Rancher, Azure DevOps, CI/CD, Git, Linux.

Platforms: Moodle LMS, Enterprise Web Applications.

EDUCATION

Master's Degree in Intelligent Computer Systems Engineering
University of Algiers 1 | 2019 – 2021

Bachelor's Degree in Computer Systems
University of Algiers 1 | 2016 – 2019

CERTIFICATIONS

- Designing RESTful APIs
- React: Software Architecture

- Unit Testing in Django
- CAL French Language Diploma – B2

LANGUAGES

- Arabic: Native
- French: B2
- English: Professional Working Proficiency